Chapter 6

Introduction to Multivariate Calculus

6.1 Why Economists are Interested

We use multivariate calculus in economics because the things we are interested in are usually functions of more than one variable. This gets suppressed in introductory economics. You may have worked with a model where the cost to a firm of producing y is $c(y) = \frac{1}{2}y^2$ and written $\frac{dc}{dy} = y$ as marginal cost. This seems a bit special, so perhaps you generalized this to work with a cost function $c(y) = \gamma y^2$. At this stage you are treating γ as a parameter of the cost function. However when you think about it the cost must depend on input prices and the technology used, so γ is itself a function, and changes in input prices change γ. You will learn to think of cost as a function of input prices $w_1,\ w_2$ and output $c(w_1, w_2, y)$. For example the minimum cost of producing output y from inputs x_1 and x_2 with prices w_1 and w_2 using the technology given by the Cobb-Douglas production function $y = Ax_1^a x_2^b$ is

$$c(w_1, w_1, y) = A^{\frac{-1}{a+b}} \left[\left(\frac{a}{b}\right)^{\frac{b}{a+b}} + \left(\frac{a}{b}\right)^{\frac{-a}{a+b}} \right] w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}}.$$

(You can find this result on page 54 of H.R.Varian, Microeconomic Analysis.) This reduces to the cost function γy^2 if $a = b = \frac{1}{4}$ and $\gamma = 2A^{-2} w_1^{\frac{1}{2}} w_2^{\frac{1}{2}}$. However, having acknowledged that costs depend on w_1, w_2 and y, you now have to use the partial derivative

$$\frac{\partial c(w_1, w_1, y)}{\partial y} = \frac{1}{a+b} A^{\frac{-1}{a+b}} \left[\left(\frac{a}{b}\right)^{\frac{b}{a+b}} + \left(\frac{a}{b}\right)^{\frac{-a}{a+b}} \right] w_1^{\frac{a}{a+b}} w_2^{\frac{b}{a+b}} y^{\frac{1}{a+b}-1}$$

for marginal cost. When you have a function of many variables $f(x_1, x_2, .x_i..x_n)$ the partial derivative is defined analogously to the derivative of a function of a single variable as

$$\frac{\partial f(x_1, x_2, .x_i..x_n)}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, .x_i + h..x_n) - f(x_1, x_2, .x_i..x_n)}{h}.$$

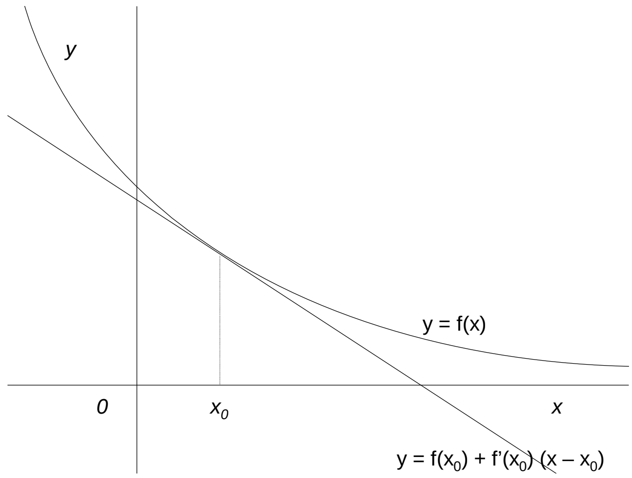

Figure 6.1: A function and its tangent

You calculate the partial derivative $\dfrac{\partial f\left(x_1, x_2, .x_i..x_n\right)}{\partial x_i}$ in exactly the same way as you calculate the derivative of a function of a single variable, treating all the variables except x_i as parameters of the function.

In fact the distinction between a parameter and a variable depends entirely on what you are interested in. Varian writes $c\left(w_1, w_1, y\right)$ treating A, a and b as parameters. But for some purposes, for example modelling technical change you might want to think of A, a and b as variables, and write costs as $g\left(A, a, b, w_1, w_2, y\right)$.

6.2 Derivatives and Approximations

6.2.1 When Can You Approximate

Think about a function of one variable $y = f\left(x\right)$ illustrated by the curved line in Figure 6.1 and its tangent at x_0 which is a straight line with equation $y = f\left(x_0\right) + f'\left(x_0\right)\left(x - x_0\right)$. The definition of the derivative $f'\left(x\right)$ implies that close to x the tangent line is a good approximation to the original function, that is

$$f\left(x\right) \approx f\left(x_0\right) + f'\left(x_0\right)\left(x - x_0\right) \text{ when } x \text{ is close to } x_0.$$

Now think about the function of two variables $y = f\left(x_1, x_2\right)$ illustrated in Figure 6.2. The graph of the function is now a curved surface. Just as the curved line in two dimensional space corresponds to a curved surface in three dimensional space, the tangent line in two dimensional space corresponds to a tangent plane in three dimensional space. In chapter 2 on vectors I showed that any plane

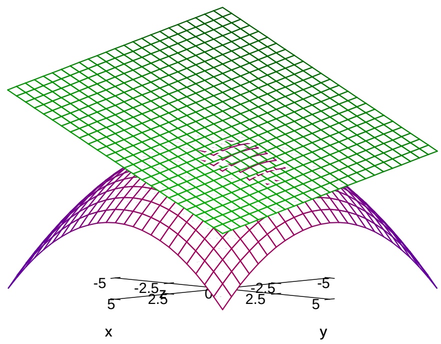

Figure 6.2: Appoximating the function at $(1,2)$ $f(x) = 100 - x_1^2 - x_2^2$ at $(1,2)$ by the plane $p(x) = 95 - 2(x_1 - 1) - 4(x_2 - 2)$.

through the point (x_{01}, x_{02}, y_0) can be written in vector notation as

$$(p_1, p_2, p_3)\left(\begin{pmatrix} x_1 \\ x_2 \\ y \end{pmatrix} - \begin{pmatrix} x_{01} \\ x_{02} \\ y_0 \end{pmatrix}\right) = 0$$

or equivalently

$$p_1(x_1 - x_{01}) + p_2(x_2 - x_{02}) + p_3(y - y_0) = 0$$

If $p_3 \neq 0$ this becomes

$$y = y_0 + a_1(x_1 - x_{01}) + a_2(x_2 - x_{02})$$

where $a_1 = -\frac{p_1}{p_3}$ and $a_2 = -\frac{p_2}{p_3}$. By analogy with the equation of the tangent line as $y = f(x_0) + f'(x_0)(x - x_0)$ the equation of the tangent plane should be

$$y = f(x_{01}, x_{02}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_1}(x_1 - x_{01}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_2}(x_2 - x_{02})$$

and the approximation should be

$$f(x_1, x_2) \approx f(x_{01}, x_{02}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_1}(x_1 - x_{01}) + \frac{\partial f(x_{01}, x_{02})}{\partial x_2}(x_2 - x_{02}).$$

This suggests that the extension to n variables should be the approximation

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}) \text{ when } \mathbf{x} \text{ is close to } \mathbf{x}_0.$$

The argument I have just given is in no way rigorous, but it can be made so. This is done by assuming that the function $f(\mathbf{x})$ can be approximated by a function of the form

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} a_i (x_i - x_{i0})$$

and then showing that a_i must be the partial derivative $\frac{\partial f(\mathbf{x}_0)}{\partial x_i}$.

You have to be slightly careful with this, because unlike the situation with functions of a single variable it is possible that the derivatives exist but the approximation does not work. As an example consider the function $f : \mathcal{R}^2 \to \mathcal{R}$ given by

$$\begin{aligned} f(x_1, x_2) &= 0 \text{ when } x_1 x_2 = 0 \\ f(x_1, x_2) &= 1 \text{ when } x_1 x_2 \neq 0. \end{aligned}$$

This function is 0 when one or both of x_1 and x_2 are zero, and 1 otherwise. Thus $f(x_1, 0) = 0$ for all values of x_1 and $f(0, x_2) = 0$ for all values of x_2, implying that the partial derivatives

$$\frac{\partial f(0,0)}{\partial x_1} = \frac{\partial f(0,0)}{\partial x_2} = 0.$$

This suggests that when (x_1, x_2) is close to $(0, 0)$

$$f(0,0) + \frac{\partial f(0,0)}{\partial x_1} x_1 + \frac{\partial f(0,0)}{\partial x_2} x_2 \tag{6.1}$$

should be a good approximation to $f(x_1, x_2)$. However

$$f(0,0) = \frac{\partial f(0,0)}{\partial x_1} = \frac{\partial f(0,0)}{\partial x_2} = 0$$

so this implies that 0 is a good approximation to $f(x_1, x_2)$ when (x_1, x_2) is close to 0. But this is very far from being a good approximation because $f(x_1, x_2) = 1$ whenever $x_1 x_2 \neq 0$.

An assumption that ensures that the approximation works for a function $f : \mathcal{R}^{++} \to \mathcal{R}$ where

$$\mathcal{R}^{++} = \{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n,\ x_i > 0 \text{ for } i = 1, 2...n\}$$

is that f has partial derivatives on $\mathcal{R}^{++}$, and that the partial derivatives are continuous. To understand what this means you need to know what continuity means. Intuitively a function $g(\mathbf{x})$ is continuous at $\mathbf{x}_0$ if $g(\mathbf{x})$ gets closer to $g(\mathbf{x}_0)$ as $\mathbf{x}$ gets closer and closer to $\mathbf{x}_0$. More formally the function is continuous at $\mathbf{x}_0$ if $g(\mathbf{x}_0)$ is the limit of $g(\mathbf{x})$ as $\mathbf{x}$ tends to $\mathbf{x}_0$.

In order to avoid writing out the conditions many times I will use the following definition.

Definition 41 *The function $f(\mathbf{x}) : \mathcal{R}^{++} \to \mathcal{R}$ is a* ***well behaved*** *if it has partial derivatives, and the partial derivatives are continuous on $\mathcal{R}^{++}$.*

If a function is well behaved the approximation

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i} (x_i - x_{i0}) \text{ when } \mathbf{x} \text{ is close to } \mathbf{x}_0$$

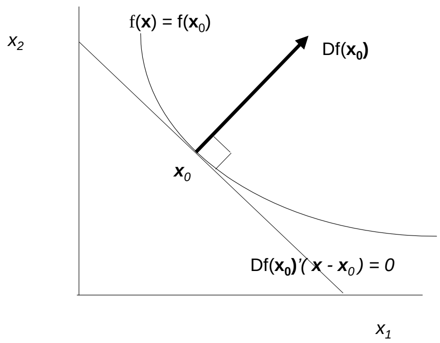

Figure 6.3: The Level Sets, Tangent and Partial Derivative Vector for a Function

works. Making this argument rigorous requires a precise definition of what is meant by approximation, and then a proof that existence and continuity of the partial derivatives implies that the approximation is valid.

The result is more general than that stated here. The set $\mathcal{R}^{++}$ can be replaced by an open subset of $\mathcal{R}^n$. See chapter, 8, but remember that this chapter is intended for MRes students and those interested in taking advanced micro (EC487). In practice economists almost always works with $\mathcal{R}^{++}$.

6.3 Level Sets and Vectors of Partial Derivatives

6.3.1 The Level Set and Tangent for One Function

A level set of a function $f(\mathbf{x})$is a set on which $f(\mathbf{x})$ is constant. Being formal about this:

Definition 42 *If S is a subset of $\mathcal{R}^n$ and the function $f : S \to \mathcal{R}$, the set*

$$\{\mathbf{x} : \mathbf{x} \in S,\ f(\mathbf{x}) = y\}$$

of elements of S for which $f(\mathbf{x}) = y$ is called a ***level set.***

A level set of a utility function is an indifference curve; a level set of a production function is an isoquant. Now think about two points $\mathbf{x}$ and $\mathbf{x}_0$ that lie in the same level set so

$$f(\mathbf{x}) = f(\mathbf{x}_0)$$

and assume that the function $f : S \to \mathcal{R}$ is well behaved so close to $\mathbf{x_0}$

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}).$$

As $f(\mathbf{x}) = f(\mathbf{x}_0)$ this implies that

$$\sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_0)}{\partial x_i}(x_i - x_{i0}) \approx 0. \tag{6.2}$$

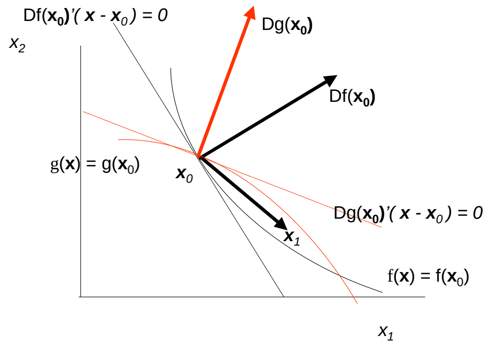

Figure 6.4: Level Sets and Partial Derivative Vectors for Two Functions

If I use notation

$$Df(\mathbf{x}_0) = \begin{pmatrix} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_2} \\ . \\ . \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_n} \end{pmatrix}$$

so $Df(\mathbf{x}_0)$ is the n vector of partial derivatives equation 6.2 becomes

$$Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) \approx 0.$$

Thus when $\mathbf{x}$ is very close to $\mathbf{x}_0$ and $f(\mathbf{x}) = f(\mathbf{x}_0)$ $\mathbf{x}$ very nearly satisfies the equation.

$$Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) = 0.$$

This is an equation of the type that I discussed at some length in section 2.7 of chapter 2 on vectors. If $n = 2$ this is a straight line, if $n = 3$ it is a plane, and for $n > 3$ it is a hyperplane. In every case the vector of partial derivatives $Df(\mathbf{x}_0)$ is orthogonal (at 90°) to the line, plane or hyperplane. This is illustrated in Figure 8.2, which shows the vector of partial derivatives $Df(\mathbf{x}_0)$ which is orthogonal to the line $Df(\mathbf{x}_0)'(\mathbf{x} - \mathbf{x}_0) = 0$. The line is a very good approximation to the level set when $\mathbf{x}$ is close to $\mathbf{x}_0$, and it seems reasonably intuitive and is in fact true that this line must be tangent to the level set.

6.3.2 Level Sets and Tangents for Two Functions

This section gives you a peek at the intuition behind Lagrangians which you will spend a lot of time studying in maths for micro and use extensively throughout your course. Figure 8.4 is the same as Figure 8.2 except that I have introduced another function $g(\mathbf{x})$, its partial derivative vector $Dg(\mathbf{x}_0)$ at $\mathbf{x}_0$ and its level set $g(\mathbf{x}) = g(\mathbf{x}_0)$. I have chosen to show the case where the two partial derivative vectors point in different directions. As the figure suggests this implies that the level sets cross each other so there is a point $\mathbf{x}_1$ which satisfies both $f(\mathbf{x}_1) > f(\mathbf{x}_0)$ and $g(\mathbf{x}_1) < g(\mathbf{x}_0)$. This tells you that $\mathbf{x}_0$ does not maximize $f(\mathbf{x})$ subject to $g(\mathbf{x}) \leq g(\mathbf{x}_0)$. The only possible solutions are those in which the two partial derivative vectors $Df(\mathbf{x}_0)$ and $Dg(\mathbf{x}_0)$ point in the same direction; this requires that $Df(\mathbf{x}_0) = \lambda Dg(\mathbf{x}_0)$ where $\lambda \geq 0$. Written out component by component this requires that

$$\begin{aligned} \frac{\partial f(\mathbf{x}_0)}{\partial x_1} &= \lambda \frac{\partial g(\mathbf{x}_0)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}_0)}{\partial x_2} &= \lambda \frac{\partial g(\mathbf{x}_0)}{\partial x_2} \end{aligned}$$

These are of course the first order conditions which come from differentiating the Lagrangian

$$\mathcal{L} = f(\mathbf{x}) + \lambda [c - g(\mathbf{x})].$$

The condition that $\lambda \geq 0$ comes from thinking about an inequality problem. You will come to think of it as the nonnegative multiplier condition of the Kuhn-Tucker theorem.